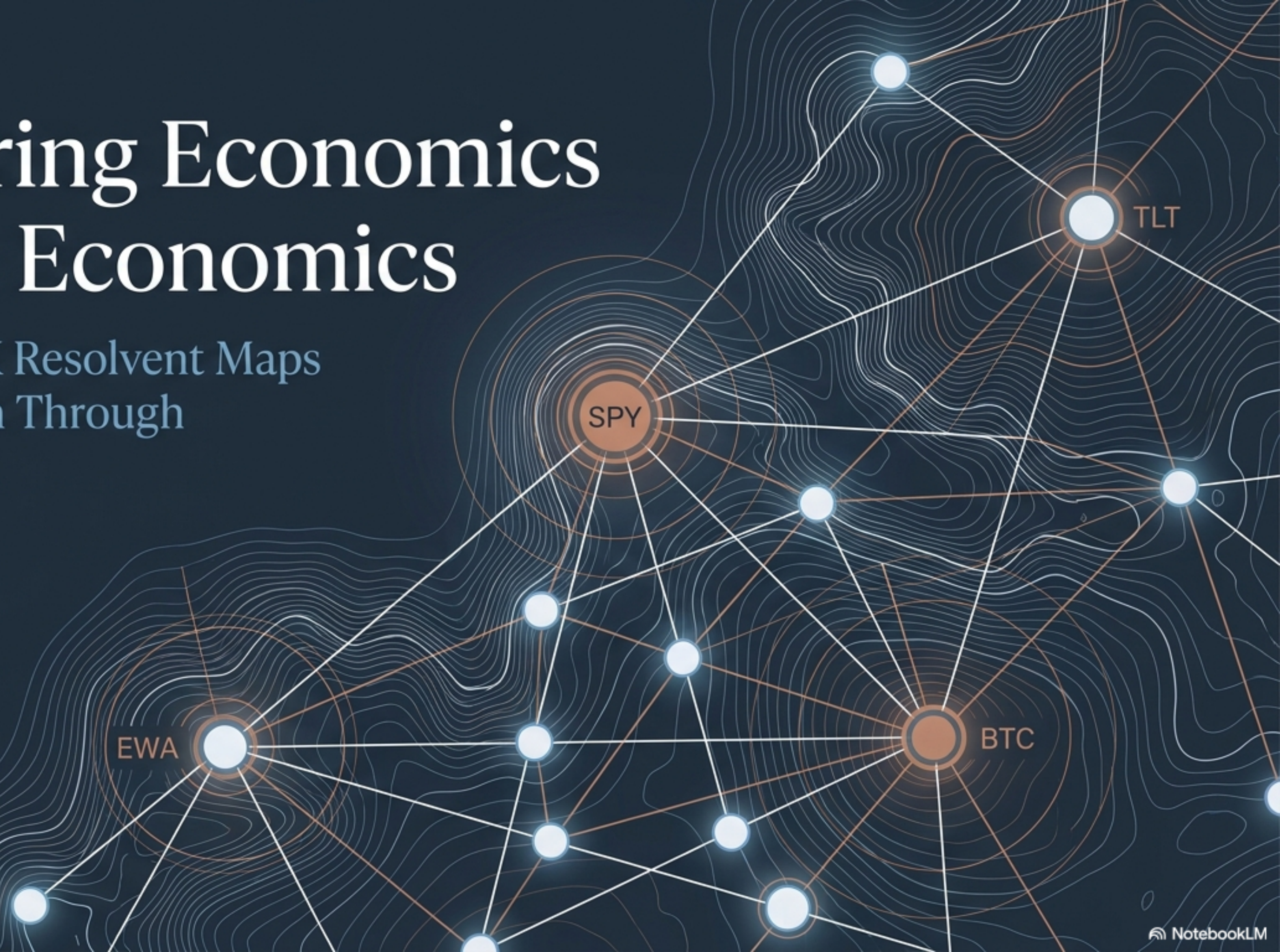
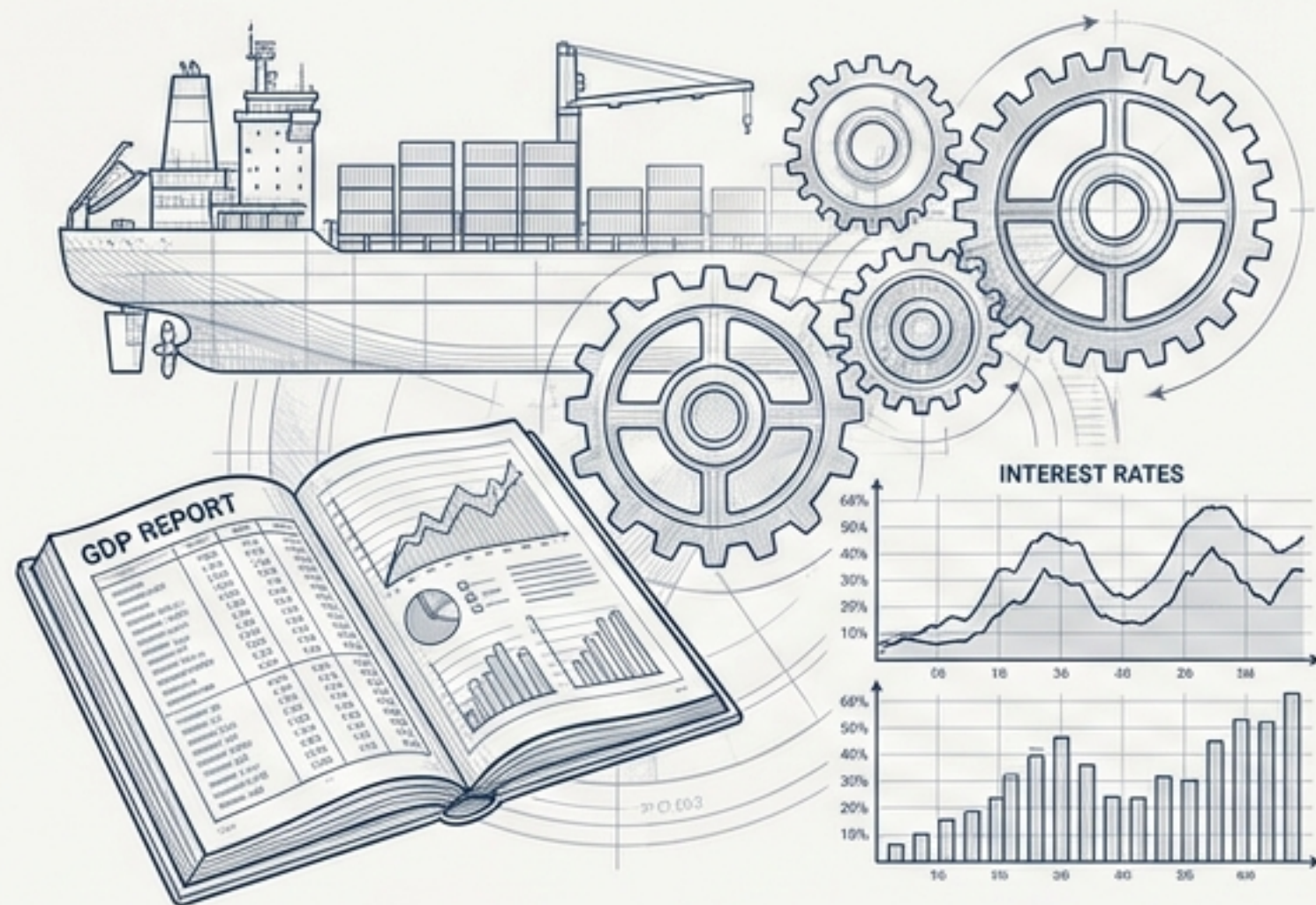


Discovering Economics Without Economics

How the MPDOK Resolvent Maps Global Contagion Through Price Topologies.

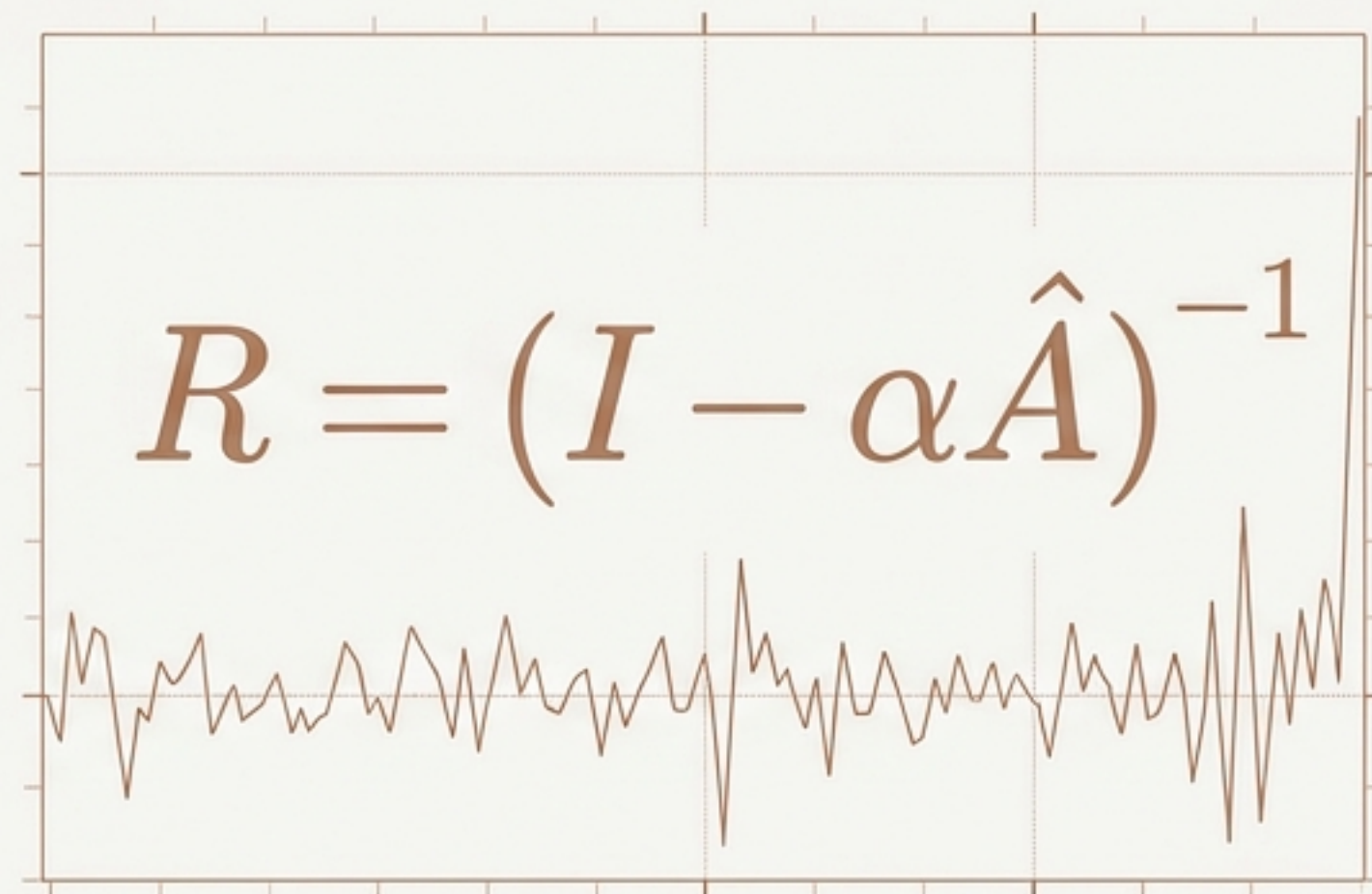


Traditional Macro Analysis



Requires heavy economic metadata:
Trade flows, GDP, rate differentials, export volumes

The MPDOK Resolvent



Requires daily price returns.
Nothing else.

The Latent Reality Thesis

The market's collective assessment of dependency is already encoded in price co-movement.
If we read the math correctly, the economics emerge automatically.

The Asset Universe

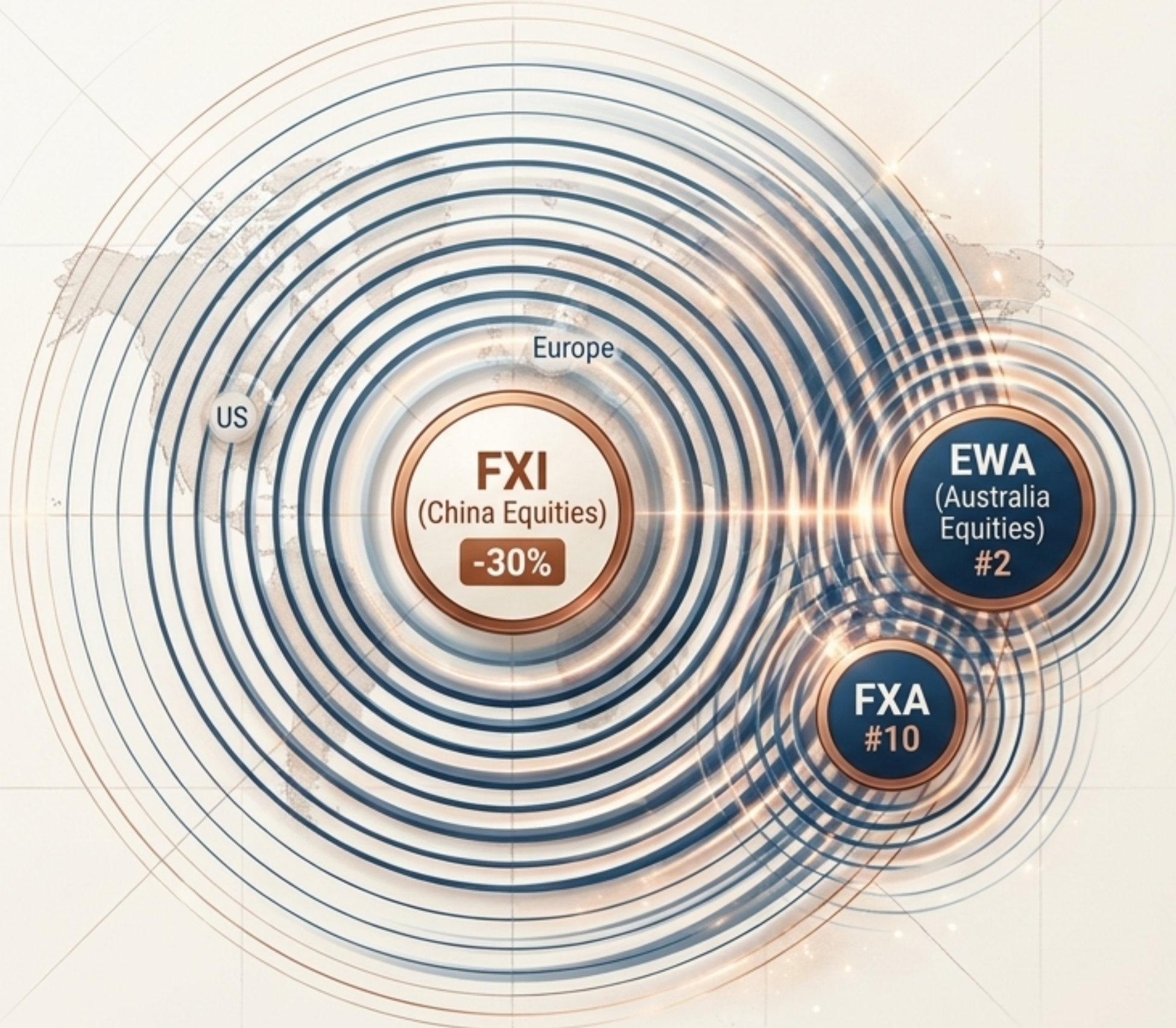
26 Nodes. 5 Classes. 1 Mathematical Engine.

Equities (10 Nodes)	Bonds (4 Nodes)	FX (5 Nodes)	Commodities (6 Nodes)	Crypto (2 Nodes)
SPY	TLT	UUP	GLD	BTC-USD
EWJ			SLV	
FXI	HYG	FXE	USO	ETH-USD
KWEB			CPER	
EWG	EMB	FXY	DBA	
EWU			UNG	
EWZ	LQD	FXA		
INDA				
EWY		CYB		
EWA				

A Universal Operator

Diverse asset classes typically require bespoke fundamental models. MPDOK applies the exact same graph resolvent to 26 nodes across 5 classes simultaneously.

Shock 1 – The Trade Channel



What the Math Found

FXI shock yields a 52.1% structural gap, hitting the theoretical α^4 floor of 52.2%. This proves China acts as a peripheral node, not an amplifying hub.

EWA (#2) ranks ahead of all other developed markets with a 2.6x ratio over $k=3$.

AUD (#10) also surfaces.

The Economic Reality: The Iron Ore Link

The model knows nothing of commodity exports or physical shipping.

Yet it correctly identifies Australia as the primary proxy for China's growth due to raw material dependence, recovering physical supply chains purely from price topologies.

Shock 2 – The Retail Behavior Channel



Target Rankings (BTC Shock)

1. BTC-USD (0.5528)

2. **EWA / Australia** (0.1142 | 2.6x)

3. SPY / US Equities (0.1118)

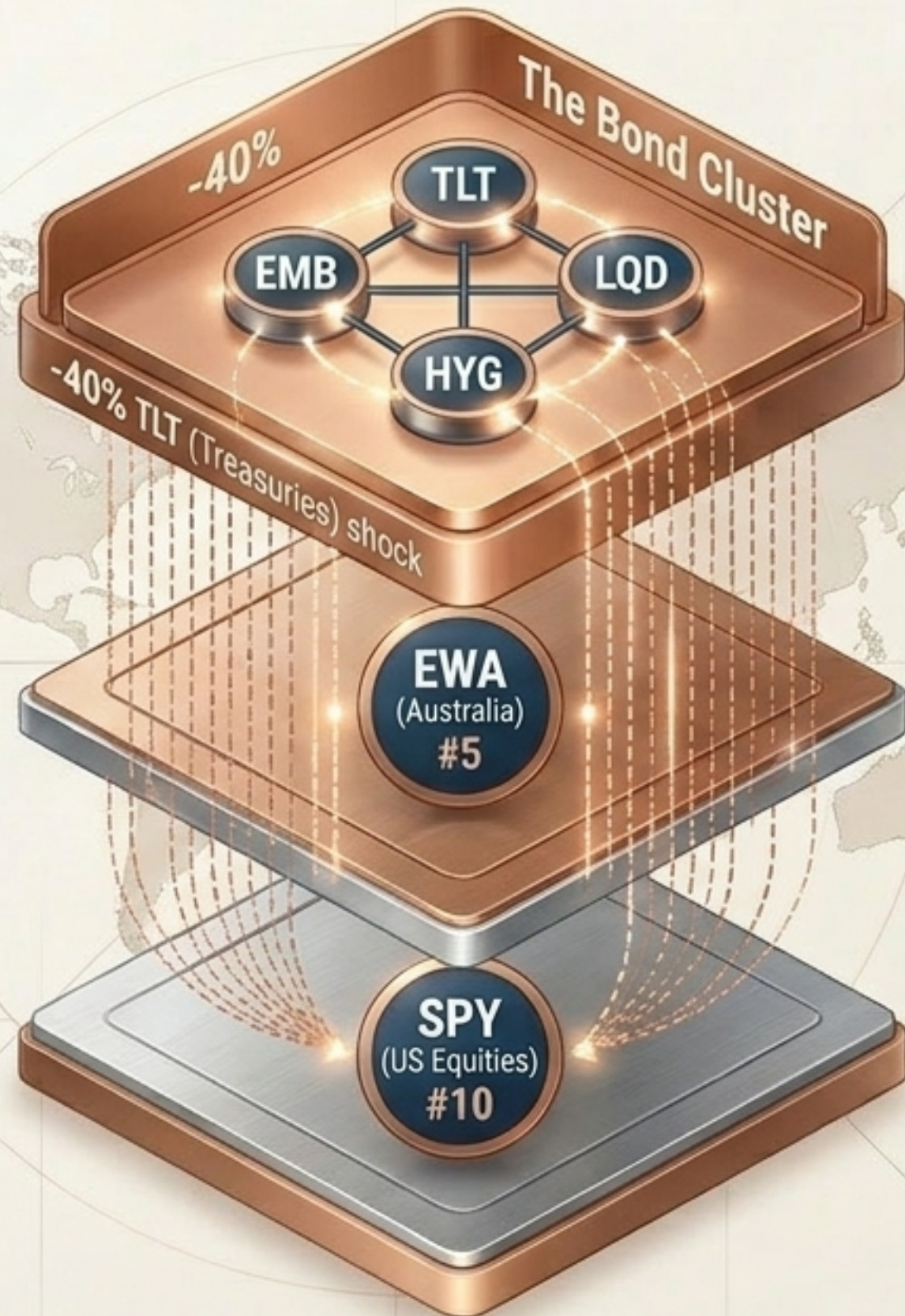
... ..

10. **EWY / Korea** (0.0977 | 2.6x)

The Economic Reality: The Retail Liquidation Effect

The resolvent traces the exact global risk-off flight-to-safety chain. Strikingly, it discovers that **Australia** and **Korea** suffer the highest relative equity damage. Without demographic surveys, the math detects the **domestic equity liquidations** required to cover **crypto** crypto margin calls in the world's highest per-capita crypto-adoption nations.

Shock 3 – The Debt Structure Channel



The Math: Spectral Diversity

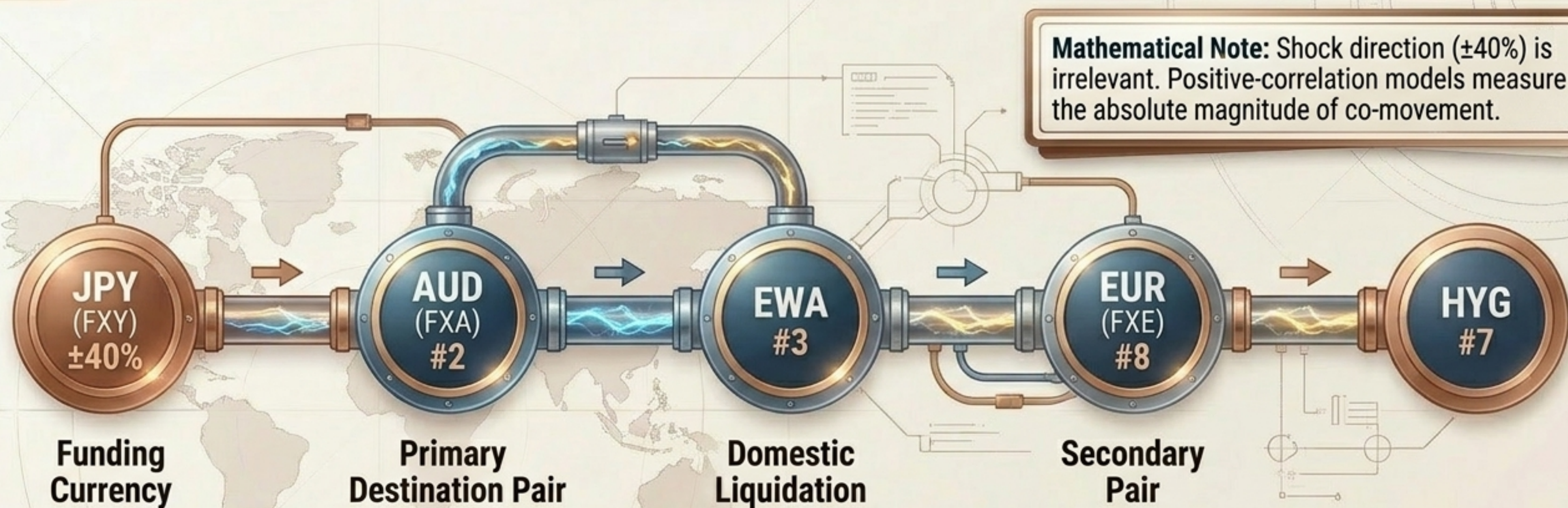
The structural gap is 45.5%, falling below the assumed 52.2% universal floor. The shock projects onto multiple eigenvectors across diverse asset classes, bypassing the worst-case degenerate hub scenario.

The Economic Reality: Variable-Rate Vulnerability

Why does a US rate shock hit Australia harder than the US?

The math detects the 2022 reality: Australian households carry immense variable-rate mortgage debt, making them hypersensitive to global yield spikes. SPY is shielded; EWA is exposed.

Shock 4 – The Financial Plumbing Channel



The Economic Reality: The Global Carry Trade

The resolvent maps the precise, multi-leg mechanics of the **Global Carry Trade** unwind. It seamlessly traces the flow from funding currency to destination pair to domestic equity liquidation—without knowing what an interest rate differential is.

The Blind Spot: Why Japan is a False Diversifier

Financial Topography

What the Positive-Only Model Sees



Hub Gap:	53.3%
Rank:	Isolated / Peripheral
Conclusion:	A perfect safe-haven diversifier.

What Happened on August 5, 2024

-12.4%
Drop

The Bank of Japan raised rates, triggering a massive yen carry unwind. The Nikkei suffered its largest drop since 1987.

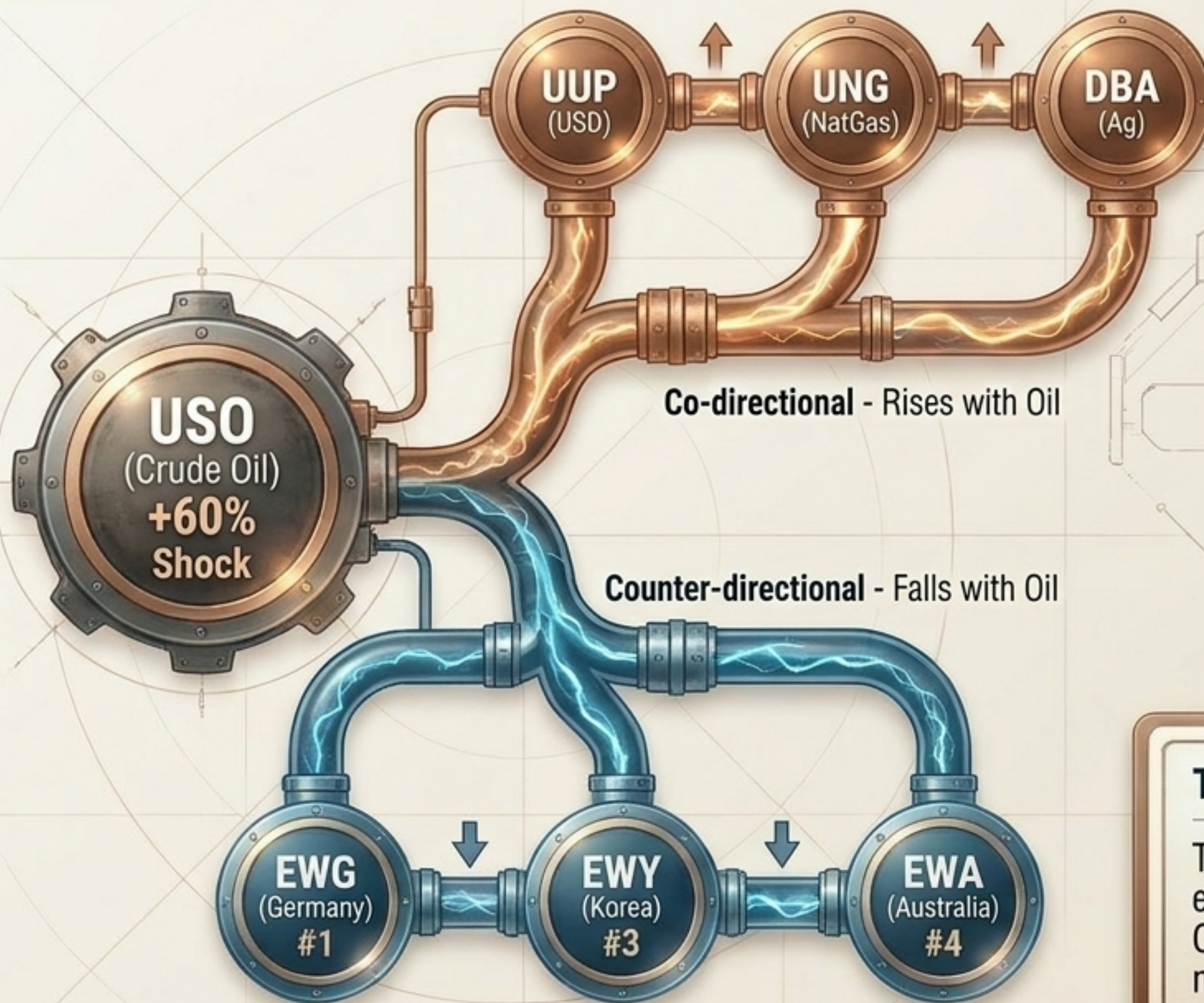
Why was the model blind to this?

Positive-only models cannot see negative correlations. The inverse relationship between the Yen and Japanese exporters zeroes out Japan's greatest risk.

We must activate the Signed Model.

The Signed Extension — The Geopolitics Channel

Financial Topography



Cancellation Risk: 9 / 10

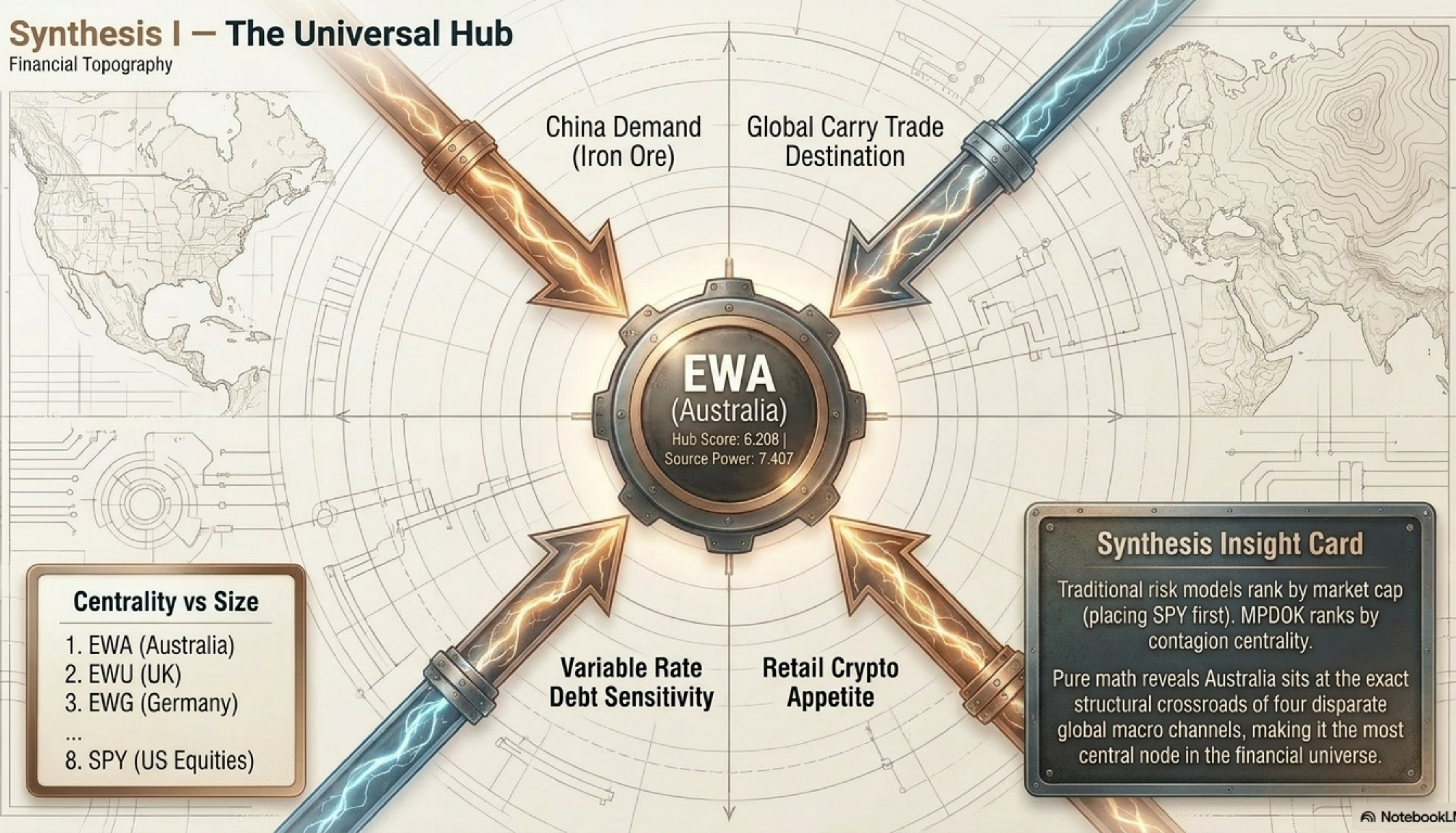
Net Impact:	+0.0982
Absolute Impact:	1.1999
Co-directional:	0.649
Counter-directional:	0.551

The Economic Reality: The Energy Importer Tax

The math perfectly maps the world's most energy-dependent economies (Germany's manufacturing, Korea's industrials). Crucially, it highlights severe Cancellation Risk: relying on net exposure completely hides massive gross bilateral risks.

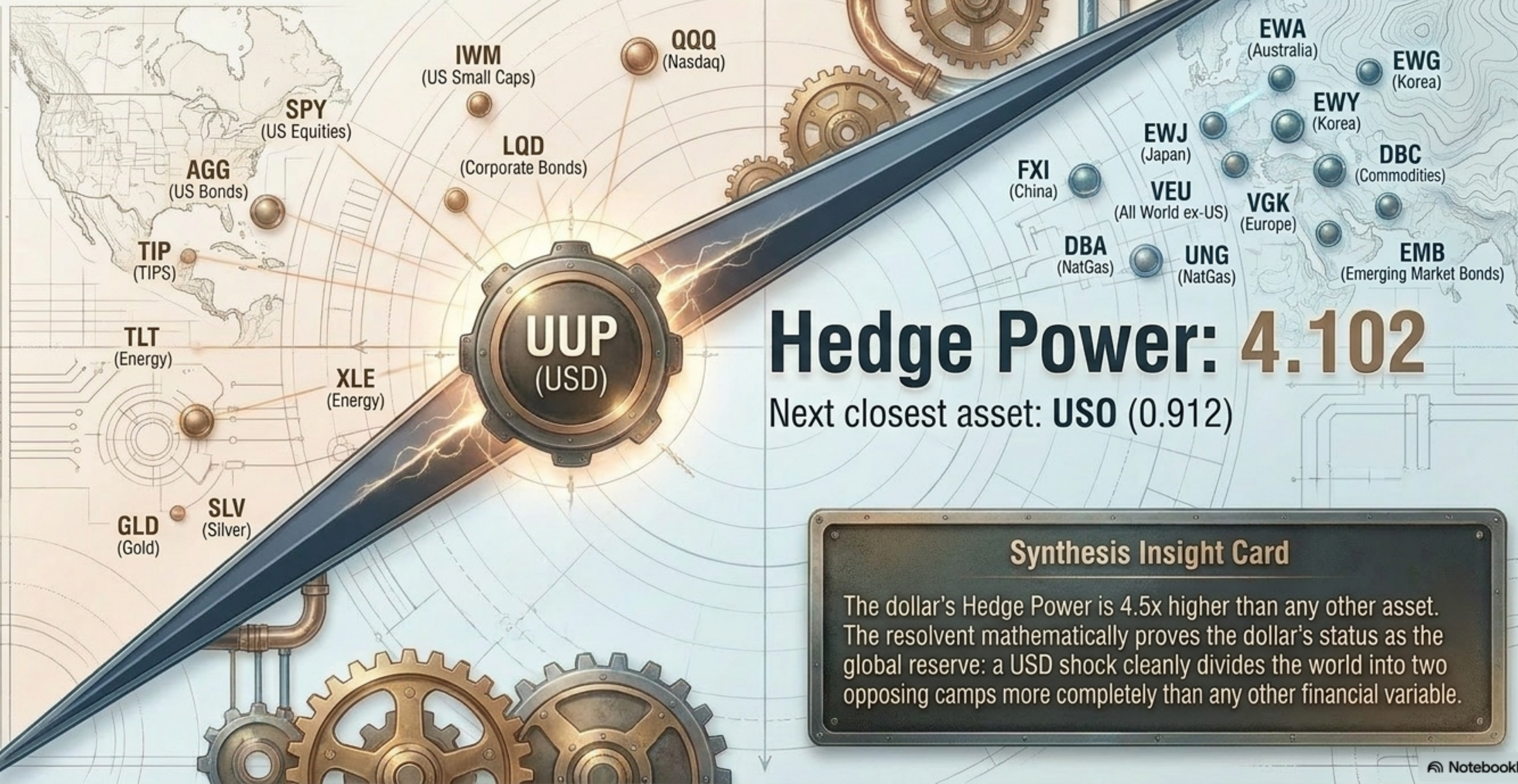
Synthesis I – The Universal Hub

Financial Topography



Synthesis II – The Reserve Currency Bifurcator

Financial Topography



Structural vs. Regime Stability

Proving mathematical laws over transient flukes (1yr, 2yr, and 3yr lookbacks)

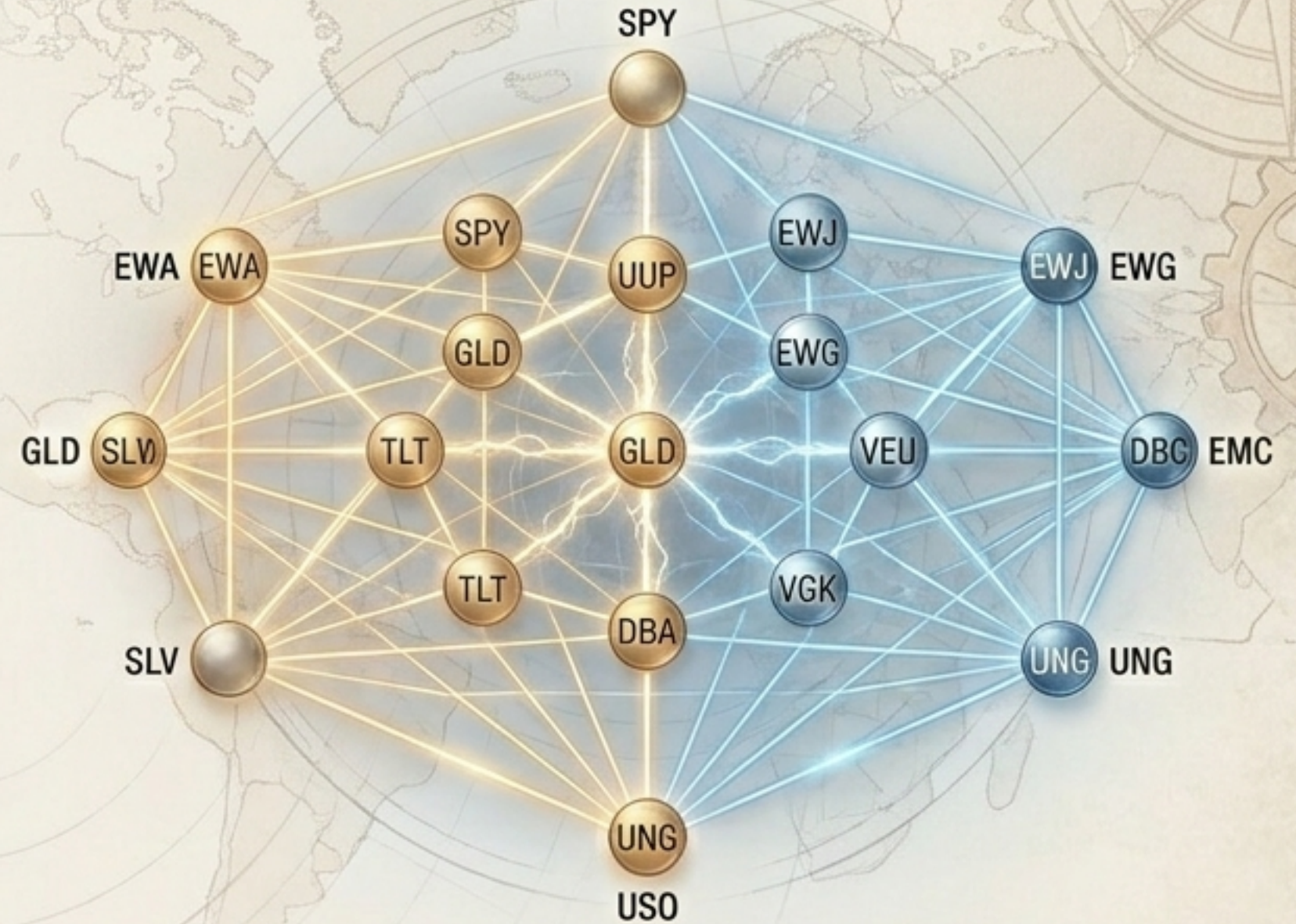


The Conceptual Model



Reinforcement Learning (AlphaGo)

An RL agent doesn't need to be programmed with the rules of strategy; it discovers the optimal structural positions from the trace of millions of games.



Graph-Resolvent Topology (MPDOK)

MPDOK operates identically in epistemology. It doesn't need economic rules pre-programmed. It discovers the optimal structure of the macro economy from the trace left by millions of daily price interactions.

Conclusion & Executive Takeaways



1. Topology Trumps Size

Market cap is a dangerous proxy for risk.

The world's largest equity market (SPY) ranks 8th in contagion centrality.

Australia (EWA) is 1st.

You must map the connections, not just the mass.



2. Beware Net Exposures

Gross bilateral exposures hide massive sector damage behind near-zero net portfolio impacts. The 9/10 cancellation risk in Oil proves that looking at net sensitivity blinds you to real economic damage.



3. The Math is Real

Price correlations contain the entirety of macroeconomic reality. Pure topological math can perfectly map physical trade routes, household debt, and global arbitrage without a single traditional economic input.